

UKSEDS COMPETITIONS

NATIONAL ROCKETRY CHAMPIONSHIP 2026-27

Rule Book

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SKYRORA

amentum

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MRC Midland Rocketry Club

Full Throttle Behaviour

Rocketry Things

WIZARD ROCKETS

UNITED KINGDOM ROCKETRY ASSOCIATION

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1. ACRONYMS & REFERENCE DOCUMENTS

1.1. Acronyms

Acronym	Description
NRC	National Rocketry Competition
RS	Requirement Specification
PDR	Preliminary Design Review
CDR	Critical Design Review
MR	Manufacturing Review
FRR	Flight Readiness Review
MRC	Midlands Rocketry Club
ECSS	European Space Agency's European Cooperation for Space Standardization
GEVS	NASA's General Environmental Verification Specification
ISO	International Standards Organization
RSO	Range Safety Officer
NOTAM	Notice To Airmen
UKRA	United Kingdom Rocketry Association
CAA	Civil Aviation Authority
COTS	Commercial Off The Shelf

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1.2. Reference Documents

1.2.1. Competition Documents

There are two other documents related to the competition, for a total of **three regulatory documents**. The three documents are as follows:

ID	Document	Description
NRC-RB	Rulebook	Outlines the overall competition rules, funding, team structures and so on - the competition logistics.
NRC-SoW	Statement of Work (This Document)	Outlines the main deliverables teams shall be expected to deliver - the competition project requirements.
NRC-RS	Requirements Specification	Outlines the technical requirements the rocket must be designed to meet

1.2.2. Useful Reference Documents

No.	Document	Description
1	ECSS-M-ST-10C Rev. 1	**Space project management** - Project planning and implementation
2	ECSS-S-ST-00-01C	**Glossary of Terms** - A list of commonly used terms in ECSS documents

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2. Overview

Once a team has designed a launch vehicle, they will have the opportunity to launch it along with all of the other teams at the NRC Competition Day. Pictures of the previous Competition Days can be found here.

3. Registration

Teams must contain a **minimum of 4 members and a maximum of 10 members** . All team members must be eligible for UKSEDS membership. More information about membership eligibility and registration (it's free!) can be found here.

Support: Finding Team Members
If you are an individual student looking to participate in NRC, but do not have access to a suitable institution to form or join a team, please email us at rocketry@ukseds.org and we will try our best to get you in touch with a local team.

UKSEDS have an extensive network of contacts at branches and in university departments across the country. We may be able to help you find team members if this is proving difficult. Please get in touch with the team contact if you need some support in organising your team.

Teams wishing to enter the competition must register. Throughout the competition, please direct any enquiries to rocketry@ukseds.org.

3.1. Large Societies Comps Liaison

Teams representing a society with more than two teams entering the competition must contact rocketry@ukseds.org before filling out the registration form. Teams must also nominate a “Competition Liaison” who shall act as a non-technical point of contact between the NRC Team and the society when required. The Competition Liaison can only be a member of one team. If the Competition Liaison is not a part of a team, they are not allowed to contribute technically to any of the society’s teams. Reports originating from the same society will be scrutinised for plagiarism. If any evidence of blatant plagiarism is found, UKSEDS reserves the right to investigate further and apply penalties.

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3.2. Motor Fees and Procurement

Note: due to substantial issues with motor supply in the UK over the last few years, **all motors for the competition shall be ordered by UKSEDS on behalf of the teams**. Furthermore, owing to the supply issues, UKSEDS cannot guarantee the availability of all Flame types, motors types or grains.

A non-refundable motor fee of £30-50 (TBC) will be charged to all teams before the preliminary design review (PDR). The exact deadline will be no later than 1 week before the PDR and will be shared with the teams well in advance. This will cover the cost of one motor per team. The motor will be made available to teams at the NRC Launch Day 2027. If a team decides to forfeit the competition, the motor will remain available for the team to use for the subsequent NRC Launch Day, their society/organisation (if either return the following year) or retained by UKSEDS as a bursary to teams requiring financial aid.

Rocket motors are more prone to failure the longer they sit on the shelf and are typically made to order. Thus, after the aforementioned one year has elapsed, the motor automatically lapses to UKSEDS ownership as a bursary for other teams. If a team misses Launch Day, UKSEDS and UKSEDS partners cannot guarantee the availability of the motors after this day.

3.3. Help From Outside Of Your Team

To ensure fair competition for all teams, the following table gives guidance on what is and is not allowed in terms of outside help.

Allowed	Not Allowed
Creation of hardware by third parties to student's design specification.	Creation of software by third parties to student's design specification.
Mentorship from outside of your team, e.g. academic supervisor or committee member, nudging your team in the right direction.	Mentorship from outside of your team, e.g. academic supervisor or committee member, telling your team what to do or managing the project.
Industry sponsorship.	Commercial off-the-shelf rocket kit.

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4. Key Dates

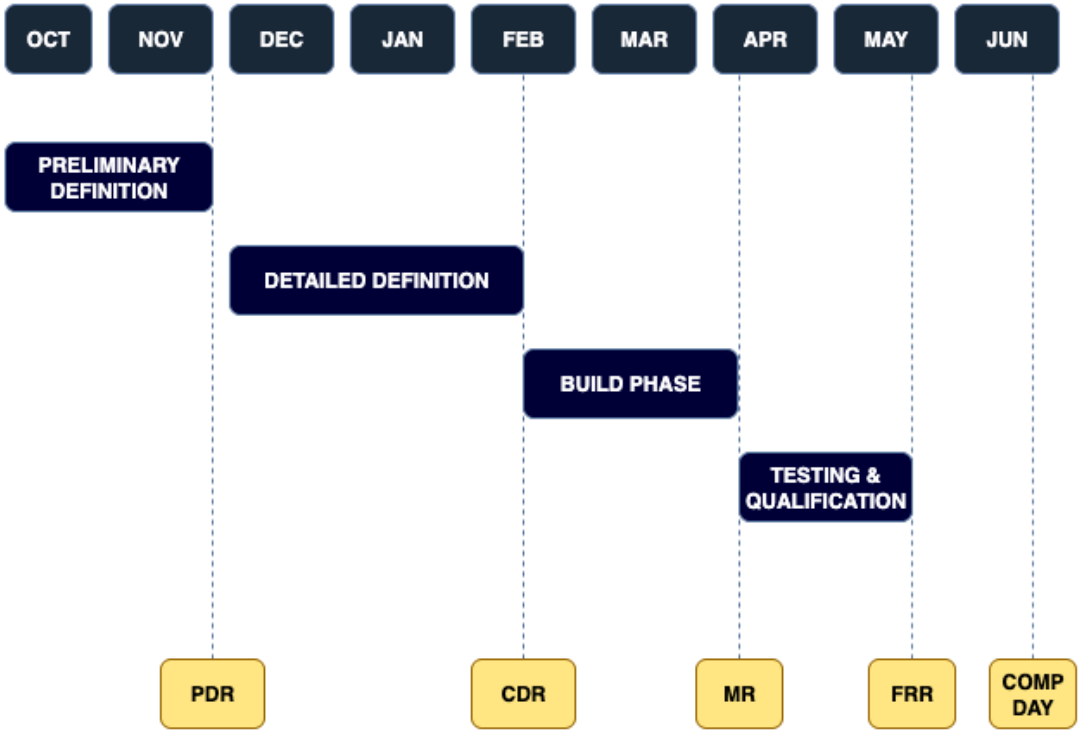


Figure 1: Key dates

4.1. Important Deadlines and Dates

Table 1: Deliverables with their respective due dates

Deliverable	Due Date
Motor Confirmation and Motor Fee	TBC
PDR (Marked)	12th December
CDR (Marked)	19th February

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MR (Marked)	10th April
FRR (Pass/Fail)	2nd June
LAUNCH DAY (Marked)	Primary Day 1: 22nd June Primary Day 2: 23rd June Backup Day 1: 29th June Backup Day 2: 30th June

5. Scoring Criteria

Teams will be scored in two main areas:

- Deliverables
- Launch Day

There is an even split in the marks between the two areas, so each team's overall mark depends on their work throughout the year just as much as their performance on Launch Day. The winning team is likely to excel in both areas.

5.1. Scoring Breakdown Table

Criteria	Points
Deliverables	600
Preliminary Design Review	200
Critical Design Review	200
Manufacturing Review	130
Monthly Reports	70
Launch Day	500
Team's preparedness	25
"The Customer's" payload integration	25
Target altitude	200
Recovery system deployment	50

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Successful recovery with minimal damage*	50
Rideshare payload data	100
Rideshare payload innovation	50
Total	**1100**

* Minimal damage shall be defined as the launch vehicle retaining its ability to be flown again on the same day with minor repairs and effectively remaining 'motor ready'.

5.2. Preliminary Design Review

Criteria	Percentage
Quality of launch vehicle design	20
Communication of design	20
Simulation data	20
Quality of rideshare payload design	10
Systems engineering	10
Project management	10
Testing	5
Media	5
Total	**100**

5.3. Critical Design Review

Criteria	Percentage
Quality of launch vehicle design	15
Communication of design	15
Simulation data	15
Quality of rideshare payload design	10

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Systems engineering	10
Project management	10
Manufacturing techniques	10
Testing	10
Media	5
Total	**100**

5.4. Manufacturing Review

Criteria	Percentage
Quality of launch vehicle manufacture	30
Communication of manufacture	20
Testing	20
Media	15
Quality of rideshare payload manufacture	10
Systems engineering	10
Project management	10
Launch vehicle video	10
Simulation data	5
Total	**130**

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5.5. Altitude Marks

A graph showing the distribution of marks against apogee altitude can be found below. Please note that heavy penalisation shall occur above 3000 ft, as this altitude is the dual-deployment limit. The Notice To Airmen (NOTAM) limit for Midlands Rocketry Club is 3500ft, so a mark of 0 shall be given to any teams that breach this limit.

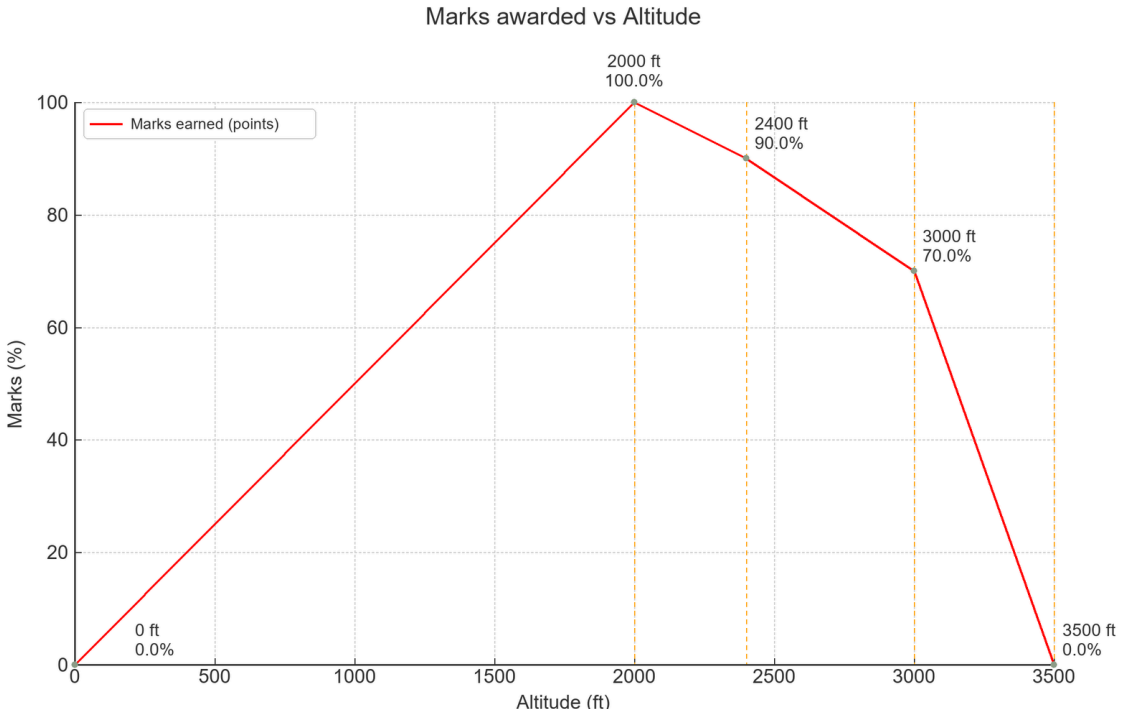


Figure 2: Distribution of marks against altitude.

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6. Deliverables

6.1. Submissions

All deliverables should be submitted through the submission portal available to teams via email. Teams should follow instructions on the specific deliverables' upload points and will be reminded by email when the submission dates are close.

6.2. Templates

All deliverables are accompanied by templates that must be used and can be found here. Additional content may be added to these templates, but the original topics covered by the templates must be included. In order to avoid penalties, teams must obey the time or page limits stated in the templates.

6.3. Deliverable Formats

The deliverable formats are as follows:

- Preliminary Design Review
  - 10-minute recorded presentation
- Critical Design Review
  - Written document, maximum of 30 pages
- Manufacturing Report
  - 10-minute recorded presentation
- Flight Readiness Review
  - Unrecorded presentation, no slide limit
- Public Engagement Submission
  - EITHER: Team choice, suggested written document or slideshow
- Monthly reports
  - Single-page written document with an optional second page of media.

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7. Rules

The UKSEDS National Rocketry Championship is organised in the spirit of healthy competition. Rules have been published below to maintain a high standard of safety and fairness for all participants.

1. This competition is open to all UK Students for the Exploration and Development of Space branches. Other teams wishing to participate are welcome but must gain consent by emailing rocketry@ukseds.org.
2. All participants must adhere to the UKSEDS Code of Conduct. The Code of Conduct is available in the NRC Statement of Work 26/27.
3. All rocketry activities must abide by the United Kingdom Rocketry Association (UKRA) Safety Code, which can be found here.
4. Submissions should be the team's own work. Any plagiarism will not be tolerated.
5. All teams and launches must abide by local laws and CAA regulations for unmanned rocket launches. Safety must take the highest priority in launch preparations and flight operations. UKSEDS retains no responsibility for the launch rules and regulations that the competing teams shall be required to follow.
6. All teams must listen to and respect the UKSEDS volunteers and the Range-Safety Officer (RSO) on the competition day. Their say is final.
7. All launches must be performed entirely through the motor's own power. Speciality launch systems (i.e. Rockoon and projectile launching) are not permitted.
8. All rockets must be successfully recovered with minimal damage. Minimal damage shall be defined as the launch vehicle retaining its ability to be flown again on the same day with minor repairs and effectively remaining 'motor ready'.. The team and the RSO must document successful recovery.
9. Team members must wait for explicit instructions from the RSO before leaving the launch pad to recover their rocket. This is to ensure health and safety as the launch site is an active airfield.

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8. Awards

As a conclusion and a commendation of the team's efforts, a limited number of awards are presented to the teams with outstanding effort in certain aspects to acknowledge their dedication. Awards are given based on various criteria based on marked deliverables throughout the year.

Award	Criteria
Winner	Awarded to the team with the best performance overall in all deliverables.
Best Rideshare Payload	Awarded to the team with the best Rideshare Payload design.
Closest to 2000ft	Awarded to the launch vehicle which achieved the closest apogee to 2000ft, as measured by "The Customer's Payload".
Public Engagement Award	Awarded to the team with the best Public Engagement.
Honorable Mentions	Awarded to the teams scoring the second and third-highest totals in all deliverables.

Additionally, there may be awards presented by sponsors on the launch days. UKSEDS reserves the right to alter any of these awards prior to the launch days, including removing them or increasing the instances of an award being presented.

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END OF DOCUMENT

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